

Indian Institute of Technology, BHU, Varanasi
Mid Term Examination (Odd Semester)- 2024-25

Date of Examination: 14-09-2023

Duration: 2 Hrs.

Full Marks: 30

Subject code: MN552

Subject: Project Management

Department: Mining Engineering

Instructions: 1. The question paper consists of 6 questions, 2. Answer any 5 questions, 3. Start each new question on a different page, 4. Calculators are allowed during the exam, 5. Marks are indicated to the right of each question. All the best!

- 1 You are provided with the following historical sales data for a product, which exhibits non-linear growth. Your task is to fit a power trend line of the form $y = at^b$, where y is the product sale number and t is the year. 6

Year	Product Sale Number
2015	100
2016	150
2017	220
2018	320
2019	460
2020	650
2021	900
2022	1250
2023	1700

- Derive the power trend line $y = at^b$ using the data provided. Show all steps clearly, including any assumptions or methods used.
- Use your derived equation to predict the sales for the year 2024.
- Calculate the coefficient of determination (R^2) for your trend line and explain how well this model fits the data.

- 2 You are provided with the following data for four Strategic Business Units (SBUs) of XYZ Corporation. Each SBU operates in a different market, with varying market share, growth rates, and competitors. Use the information given to calculate the relative market share for each SBU, categorize them according to the BCG Matrix, and determine which SBU, if any, should be divested. 6

SBU	Market Share (%)	Closest Rival	Rival's Market Share (%)	Market Growth Rate (%)
SBU 1 (TechGiant)	60	Competitor A	50	18
SBU 2 (GreenEnergy)	70	Competitor B	35	3
SBU 3 (EcoFood)	25	Competitor C	45	22
SBU 4 (CityTransport)	15	Competitor D	30	1

- Calculate the relative market share for each SBU.
- Categorize each SBU into the appropriate quadrant of the BCG Matrix based on relative market share and market growth.
- Recommend which SBU, if any, should be divested, providing a justification for your decision.

3 What key aspects are considered in technical analysis? Discuss each with relevant examples.

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4 a. Using the following data, prepare a Profit and Loss Statement for the company:

Expected Sales: ₹50,00,000	Financial Expenses: ₹1,00,000
Cost of Production: ₹30,00,000	Depreciation: ₹2,00,000
Administrative Expenses: ₹5,00,00	Other Income: ₹50,000
Sales Expenses: ₹2,00,000	Provision for Taxation: 30% of Profit Before Taxation
Royalty and Know-how Payable: ₹3,00,000	Dividend Payable: ₹2,00,000

1. Prepare a **Profit and Loss Statement** using the provided data.
2. Include all calculations to determine total costs, gross profit, operating profit, profit before taxation, provision for taxation, profit after tax, and profit after dividend payment.
3. Calculate the **Net Cash Accrual** by including depreciation in the final retained profit.

b. Discuss the contents of a balance sheet.

5 a. Mr. Vinay plans to send his son for higher studies abroad after 10 years. He expects the cost of these studies to be ₹10,00,000. How much should he save annually to have a sum of ₹10,00,000 at the end of 10 years, if the interest rate is 12 percent?

b. A company is considering investing ₹5,00,000 in a project that is expected to generate cash flows of ₹1,50,000 per year. Calculate the **payback period** for the project and determine whether the project should be accepted if the company's maximum acceptable payback period is 4 years.

6 a. Briefly describe the phases of capital budgeting and explain the key activities involved in each phase.

b. You are planning to save for a vacation that costs ₹50,000 in 3 years. If your savings account offers an annual interest rate of 4%, how much do you need to deposit today (the present value) to reach your goal of ₹50,000?

Important Formulas

- $\beta_1 = (\sum (x_i - \bar{x})(y_i - \bar{y})) / (\sum (x_i - \bar{x})^2)$
- $\beta_0 = \bar{y} - \beta_1 \cdot \bar{x}$
- $R^2 = 1 - (SS_{res} / SS_{tot})$
- $SS_{tot} = \sum (Y_i - \bar{Y})^2$
- $SS_{res} = \sum (Y_i - \hat{Y}_i)^2$
- $FV_n = PV \cdot (1 + r)^n$
- $PV = FV_n / (1 + r)^n$
- $FV_{A_n} = A \cdot [((1 + r)^n - 1) / r]$
- $PV_{A_n} = A \cdot [1 - (1 + r)^{-n}] / r$

*****End of Question Paper*****